Long-term Maturation of Congenital Diaphragmatic Hernia Treatment Results

Toward Development of a Severity-Specific Treatment Algorithm

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Objectives: To assess the impact of varying approaches to congenital diaphragmatic hernia (CDH) repair timing on survival and need for ECMO when controlled for anatomic and physiologic disease severity in a large consecutive series of patients with CDH.

Background: Our publication of 60 consecutive patients with CDH in 1999 showed that survival was significantly improved by limiting lung inflation pressures and eliminating hyperventilation.

Methods: We retrospectively reviewed 268 consecutive patients with CDH, combining 208 new patients with the 60 previously reported. Management and ventilator strategy were highly consistent throughout. Varying approaches to surgical timing were applied as the series matured.

Results: Patients with anatomically less severe left liver-down CDH had significantly increased need for ECMO if repaired in the first 48 hours, whereas patients with more severe left liver-up CDH survived at a higher rate when repair was performed before ECMO. Overall survival of 268 patients was 78%. Survival was 88% for those without lethal associated anomalies. Of these, 99% of left liver-down CDH survived, 91% of right CDH survived, and 76% of left liver-up CDH survived.

Conclusions: This study shows that patients with anatomically less severe CDH benefit from delayed surgery whereas patients with anatomically more severe CDH may benefit from a more aggressive surgical approach. These findings show that patients respond differently across the CDH anatomic severity spectrum and lay the foundation for the development of risk-specific treatment protocols for patients with CDH.

Keywords: CDH, congenital diaphragmatic hernia, ECMO, outcomes, repair, severity, survival, timing

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Congenital diaphragmatic hernia (CDH) is a common and frequently lethal birth defect that has occupied the hearts and minds of pediatric surgeons, neonatologists, and devastated parents for decades. Currently, the best estimate of overall survival is 67%, as reported by the CDH Study Group, which is made up of centers interested in this disease and which voluntarily report their results.[1] The real survival rate for CDH may be well less than this, but conservatively at least 1 in 3 infants born with CDH do not survive.

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This is a single-center, retrospective review of a 19-year experience in congenital diaphragmatic hernia.

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Improvement in survival has occurred over the last 2 decades, most likely due to the wide permeation of techniques of gentle ventilatory support for these patients, pioneered by Wung et al[2] and first reported for CDH in the 1990s.[3–5] These concepts have almost universally been successful in improving outcomes in centers that adopt them.[6–10] Adoption of gentle ventilatory techniques is usually paired with the adoption of delayed repair of the CDH.[3,4,11,12] Although the pertinent literature strongly supports the role of gentle ventilation in CDH, the case for survival benefit specifically attributed to delayed repair of CDH is missing, especially as any effect is difficult to separate from the effect of improved ventilatory techniques.[12–14] Although it is well known that anatomic markers of CDH severity exist with correspondingly different survival rates,[15–17] previous reports evaluating the effects of surgical timing on CDH survival have not addressed these differences.

In 1992, we adopted the gentle ventilation strategies of Wung and Stolar,[3] and published our first 60 patients in 1999.[5] The notable survival rates in that series, 78% overall and 89% in patients without associated lethal anomalies, resulted in the development of an expanded referral practice to our center. Since our original report, we have accrued an additional 208 consecutive patients, now totaling 268.

We postulated that patients with more severe CDH face different physiologic challenges from patients with less severe CDH and might respond differently to varying treatment strategies. The gentle ventilation strategies we originally reported remained highly consistent in this series but we applied varying approaches to surgical timing. We critically review this 19-year experience to see how differences in surgical timing may have affected survival and need for ECMO across the spectrum of CDH severity.

METHODS

All newborns with Bochdalek CDH, symptomatic in the first 6 hours of life and cared for at the University of Florida and Shands Hospital for Children between September 1992 and December 31, 2011 were included, regardless of associated anomalies, degree of pulmonary hypoplasia, and medical condition on arrival. Two separate hospital medical record queries were cross-referenced with operative records, autopsy records, a pediatric surgery database, and 2 prenatal evaluation databases to ensure that no patients were missed. Patients with Morgagni CDH, diaphragmatic eventration, and patients in whom diagnosis of diaphragmatic hernia was delayed for more than 48 hours after delivery were not included. This study was approved by the University of Florida Institutional Review Board.

Type of Study

A retrospective review was performed. Study data were collected and managed using REDCap electronic data capture tools hosted at the University of Florida and supported by the University of Florida Clinical and Translational Science Institute. Data collected and used for this analysis include gestational age, birth weight, Apgar score at 1 minute, Apgar score at 5 minutes, predicted survival,[18] date, time, and place of delivery, side of defect, stomach position, liver position, presence of associated anomalies, and surgical details including patch. Anatomic findings were reported from direct observation at surgical repair, autopsy, or prenatal imaging if neither was available. Use and type of ECMO, duration of ECMO, and condition at discharge were also collected. Time of CDH repair and of ECMO initiation were collected and analyzed. Total days on ECMO and number of ECMO runs were recorded. Patients who received ECMO before repair of CDH were judged to have had "opportunity" for repair if they had preductal saturations above 90% and adequate hemodynamic stability for 16 hours after birth for surgical repair to have been possible, had it been chosen. Those who did not meet these criteria were judged to "not have opportunity" for repair before ECMO. CDH Study Group predicted survival was retrospectively calculated from http://nicutools.org.

Evaluation and Treatment

Patients referred for prenatal evaluation were counseled optimistically and terminations for isolated CDH did not occur. Delivery was planned between 38 and 39 weeks, with vaginal delivery preferred. EXIT to ECMO was not used. The pediatric surgeon attended the deliveries and prenatally diagnosed patients were intubated immediately. Apgar scores were independently assigned by the neonatal staff. Ventilator strategy, blood gas goals, fluid management, and hemodynamic support have been previously published[5] and did not change appreciably. The attending pediatric surgeon provided management oversight throughout the hospitalization.

Nitric oxide was used as first-line rescue for critical instability (oxygenation index > 40) and ECMO was used only for persistent or recurrent critical instability (oxygenation index >40) and only after optimization of ventilation and medical support, including use of pressors, steroids, and other vasoactive medications as deemed indicated.

Management strategies regarding timing of CDH repair and management of repair decisions related to ECMO evolved through the 19-year experience. As no convincing data existed showing superiority of any surgical timing strategy,[12] a variety of timing strategies were employed over the series and provide significant variation for analysis. Treatment difficulties and survival trends appreciated during the series resulted in ongoing evolution of the repair paradigm. Delay of repair beyond 48 hours was the prevalent strategy, but repairs before this did occur, sometimes for specific clinical reasons, and sometimes not. Delay of repair in patients with more severe CDH resulted in the majority of those patients arriving to ECMO unrepaired, and several such patients we thought should have survived suffered poor outcomes. To avoid the situation of arriving to ECMO unrepaired, we slowly evolved a more aggressive approach to surgical repair in patients we deemed likely to need ECMO. This conclusion was based on prenatal and postnatal data such as lung-to-head ratio (LHR) liver position, and blood gas trends. This analysis was designed to define which repair strategies worked best and for which patients.

For patients who arrived to ECMO unrepaired, strategies for repair also evolved. Earlier in the series, these patients were generally repaired on ECMO, whereas later in the course attempts were made to stratify those patients as eventually weanable from ECMO or not, judged on a clinical estimate of pulmonary parenchymal volume. Patients judged to have "more" lung parenchyma on the basis of clinical and radiographic criteria were treated with the goal of repair after ECMO, whereas patients judged to have critically small lungs were repaired on ECMO, usually early in the course. Surgical technique and hemostasis were exacting, and aminocaproic acid (Amicar) was not used routinely.

Results were analyzed for the 240 patients without lethal associated anomalies in aggregate and stratified by the anatomic cohorts of left liver-down (LLD), right, and left liver-up (LLU) CDH. Survival to discharge and need for ECMO were compared by univariate and multivariate techniques, looking specifically at risk for ECMO related to surgical timing, and survival on the basis of the order relationship of surgical repair to ECMO. Patients who did not have "opportunity for repair" before ECMO were not included in the comparison of survival outcomes related to order of surgical repair and ECMO. Prenatal evaluation data such as LHR were available for the majority of patients but were missing in enough patients that these data were excluded from this analysis.

Statistical Analysis

Unless otherwise specifically noted, comparison of continuous cohorted variables was performed using 2-tailed t test, and exact analysis of proportion was performed using Fisher exact test. Quantitative variables are expressed as the mean ± standard deviation. All tests were 2-sided and the threshold for statistical significance was set to $P < 0.05$. We used logistic regression to evaluate the effect of ECMO first versus surgery first on survival (yes or no) when controlling for CDH Study Group–predicted survival.[18] We also used multivariate logistic regression to evaluate the effect of timing of surgery (within 48 hours vs after 48 hours) on the need for ECMO when controlling for predicted survival. The distributions of survival time were estimated using the adjusted Kaplan-Meier method and compared using a log-rank test. Multivariate Cox proportional hazards models were used to assess prognostic and treatment factors for overall survival and to compute hazard ratios and their 95% confidence intervals. The ability to do multivariate analysis of anatomic subgroup outcomes as it related to surgical decisions was limited by the low number of events (deaths) in the anatomic subgroups. Only a single added variable was statistically appropriate for multivariate analysis, and predicted survival as calculated from the CDH Study Group, which fits the model well (Table 1), was used. Analyses were performed with R statistical software (version 2.15.0; R foundation, Vienna, Austria).

RESULTS

A total of 268 patients with CDH were identified. Sixty of these were previously reported[5] and 208 are new. Overall survival to discharge was 210 patients (78%). The CDH Study Group–predicted survival for this group on the basis of birth weight and 5-minute Apgar score was 59.8%, $P < 0.001$. All patients were discharged home breathing spontaneously, without surgical airways, and on no support other than nasal cannula oxygen.

A total of 28 patients had associated anomalies that we deemed lethal or highly severe and which either independently or in combination with the CDH had a devastating effect on the patient's chance for survival (Table 2). The majority of these patients were diagnosed prenatally and were not treated with intent to cure. Five were treated, did not survive, and are included here because of the severity of the associated anomaly, which if included would confound the treatment analysis. These 28 patients are excluded from subsequent analysis, which follows.

Two hundred forty patients without lethal associated anomalies were encountered. Prematurity of 34 weeks or less affected 30 patients (30 of 240, 12%) and patients with tetralogy of Fallot (n = 2), coarctation of the aorta (n = 2), ventricular septal defect (n = 18), and atrial septal defect (n = 15) are included, as are a number of patients with nonlethal chromosomal inversions, additions, and deletions. Two patients with congenital cytomegalovirus, both of whom died and diagnosis of cytomegalovirus was made at autopsy, are also included in the treatment analysis. Two hundred ten of 240 patients (88%) without associated lethal anomalies survived.

TABLE 1. Anatomic Subgroups (Lethal Associated Anomalies Excluded)

	Left Liver-Down (n = 97)	Right (n = 42)	Left Liver-Up (n = 101)	*P*
Gestational age				
Mean ± SD	37.4 ± 2.1	36.9 ± 2.8	37.2 ± 2.1	*P* = 0.321*
Median (IQR)	38 (37–39)	37 (36–38)	38 (36–38)	
Birth weight				
Mean ± SD	3022 ± 582	2926 ± 550	2769 ± 680	*P* = 0.0169*
Median (IQR)	3117 (2650–3440)	2984 (2563–3210)	2840 (2400–3175)	
Apgar-1				
Mean ± SD	5.26 ± 2.27	3.90 ± 2.15	3.52 ± 2.24	*P* < 0.0001*
Median (IQR)	5 (4–7)	4 (2–6)	3 (1–5)	
Apgar-5				
Mean ± SD	7.60 ± 1.59	6.37 ± 2.07	5.69 ± 2.18	*P* < 0.0001*
Median (IQR)	8 (7–9)	6 (5–8)	6 (4–7)	
CDH SG- predicted survival				
Mean ± SD	73.8% ± 16.4%	61.7% ± 18.8%	52.6% ± 25.0%	*P* < 0.0001†
Median (IQR)	79% (66%–86%)	64% (50%–78%)	54% (38%–75%)	
Survival, n (%)	96 (99.0%)	38 (90.5%)	76 (75.2%)	*P* < 0.0001‡

N = 240. *P* values are the result of the following:
*Analysis of variance, with pairwise comparisons adjusted using Tukey's method for multiple comparisons.
†Kruskal-Wallis test, with pairwise comparisons adjusted using Bonferroni's method for multiple comparisons.
‡Chi-square test, with pairwise comparisons adjusted using Bonferroni's method for multiple comparisons.
IQR indicates interquartile range (the 25th and 75th percentiles).

TABLE 2. Lethal or Highly Severe Associated Anomalies (28 of 268, 10%) (Removed From Subsequent Treatment Analysis)

Chromosomal	4
Trisomy 12	(1)
Trisomy 18	(1)
Tetrasomy 12p	(2)
Cardiac	8
Hypoplastic L heart	(2)
DORV	(1)
Transposition, single ventricle	(1)
Truncus arteriosus	(1)
Tricuspid atresia	(1)
Complete heart block	(1)
Noncompaction cardiomyopathy	(1)
CNS malformations	3
Iniencephaly	(1)
Serious brain malformation with hydrocephalus. Rx withheld	(2)
Multiple combined	5
Cloacal exstrophy, myelomeningocele	(1)
Hepatopulmonary fusion, AV canal	(1)
Combined severe vertebral and chest wall deformity	(3)
Bilateral CDH	6
Bilateral	(5)
Bilateral with DORV	(1)
DOA (outborn, arrived with/without vitals, CPR in progress, pupils fixed)	2

AV indicates atrioventricular; CNS, central nervous system; CPR, cardiopulmonary resuscitation; DOA, dead on arrival; DORV, double outlet right ventricle.

Anatomic Subsets

Mean birth weight, Apgar-1, Apgar-5, predicted survival, and observed survival all declined significantly across the CDH spectrum, with LLD the least severe, right CDH was intermediate, and LLU CDH the most severe (Table 1). Details of treatment related to repair, ECMO, and survival outcomes are presented in Table 3.

Analysis of Survival Related to Timing of Repair Relative to ECMO

Ninety-six patients were treated with ECMO. Twenty-seven of these went on ECMO in the first 16 hours and were not considered to have opportunity for repair. Sixty-nine patients required ECMO and had opportunity for repair before ECMO. Twenty-seven of 30 patients repaired first, followed by ECMO survived (90%) and 28 of 39 patients treated with ECMO first survived (72%).

Combined

Kaplan-Meier survival analysis ($P = 0.0656$), multivariate logistic regression controlling for predicted survival ($P = 0.132$), and Cox proportional hazards model ($P = 0.1196$) showed a trend to improved survival in patients repaired before ECMO but this did not achieve statistical significance.

Left Liver-Down

Only 11 of 97 LLD patients required ECMO and only 1 died, severely limiting statistical analysis of this subgroup. Univariate analysis ($P = 0.455$) and Kaplan-Meier survival analysis ($P = 0.273$) showed no statistically significant relationship between surgical repair timing relative to ECMO, and survival.

Right

Forty-two patients with right CDH were treated and 23 required ECMO. Univariate analysis ($P = 0.525$) and Kaplan-Meier survival analysis ($P = 0.307$) showed no significant difference in outcomes in patients repaired before ECMO versus those who arrived on ECMO unrepaired.

Left Liver-Up

Forty-three LLU patients who required ECMO had opportunity for repair. Eighteen were treated with repair first and 17 survived (94%), whereas 25 arrived on ECMO unrepaired and 16 survived (64%, $P = 0.028$). Kaplan-Meier survival analysis is shown in Figure 1. Mean predicted survival in the patients who underwent repair first was 59.3 ± 24%, whereas the mean predicted survival in patients treated with ECMO first was 52.8% ± 19%. Multivariate

TABLE 3. Anatomic Subgroups and Outcomes

CDH (Lethal Associated Anomalies Removed)	Total	Left Liver-Down	Right	Left Liver-Up
No. patients	240	97	42	101
Survived (%)	210 (88)	96 (99)	38 (91)	76 (75)
No. non-ECMO	144	86	19	39
Mean time to repair (hr)	106 ± 71	106 ± 77	87 ± 44	117 ± 68
Survived (%)	138 (96)	86 (100)	19 (100)	33 (85)
No. ECMO (%)	96 (40)	11 (11)	23 (55)	62 (61)
Survived (%)	72 (75)	10 (91)	19 (83)	43 (69)
No. ECMO in 1st 16 hr of life	27	0	8	19
Survived (%)	17 (63)	—	7 (88)	10 (53)
No. ECMO patients with "opportunity for repair"	69	11	15	43
No. ECMO patients repaired first, ECMO second	30	6	6	18
Mean time to repair (hr)	40.9 ± 43	27 ± 15	35 ± 26	48 ± 53
Survived (%)	27 (90)	6 (100)	4 (67)	17 (94)
No. ECMO patients. ECMO first	39	5	9	25
Mean time to ECMO (hr)	45 ± 40	31 ± 10*	34 ± 16	47 ± 42
Survived (%)	28 (72)	4 (80)	8 (89)	16 (64)
All ECMO firsts, repair on ECMO	34 of 66	0 of 5	12 of 17	22 of 44
Survived (%)	23 (68)	—	12 (100)	11 (50)
All ECMO firsts, repair after ECMO	25 of 66	4 of 5	4 of 17	17 of 44
Survived (%)	22 (88)	4 (100)	3 (75)	15 (88)
All ECMO firsts, did not achieve repair	7 of 66	1 of 5	1 of 17	5 of 44
Survived (%)	0	0	0	0
Survival comparison repair firsts versus ECMO first (controlled for CDH SG–predicted survival)	$P = 0.132$	$P =$ NS	$P =$ NS	$P = 0.0549$

CDH SG indicates CDH Study Group; ECMO, extracorporeal membrane oxygenation; NS, not significant.

logistic regression controlling for predicted survival (Fig. 2) showed a strong trend to improved survival with repair first that failed to meet accepted standards for statistical significance, $P = 0.0549$. Cox proportional hazards modeling showed that survival was marginally better in the repair-first group ($P = 0.067$). It is estimated by this model that LLU patients who were repaired before ECMO faced 0.14 times the hazard of death as ECMO-first LLU patients (95% confidence interval = 0.017–1.15).

Risk of ECMO

The majority of risk of ECMO occurred in the first 48 hours in LLD patients and in the first 96 hours in LLU patients. We calculated the incidence of ECMO in patients who underwent repair in the first and second 48-hour intervals after birth and compared these with patients not repaired during these periods and who were not already repaired or on ECMO.

Combined

Timing of surgery proved to be a significant predictor of ECMO in patients with CDH. Multivariate logistic regression controlled for predicted survival estimated that patients repaired within the first 48 hours of life have 2.5 times the odds of ECMO compared with patients repaired later or not at all (95% confidence interval = 1.23–5.13), $P = 0.0113$. CDH Study Group–predicted survival is a significant predictor of needing ECMO, $P < 0.0001$.

Left Liver-Down

Six of 21 patients with LLD CDH repaired in the first 48 hours (29%) went to ECMO whereas only 5 of 71 (7%) patients not repaired during this time interval went to ECMO (Fig. 3). Multivariate logistic regression controlling for predicted survival estimated a 6.3-fold increase in risk of needing ECMO in patients with LLD CDH repaired in the first 48 hours ($P = 0.0110$).

Right

Five of 9 (55%) patients with right CDH repaired in the first 48 hours went to ECMO whereas 10 of 24 (42%) not repaired in the first 24 hours went to ECMO. One of 8 repaired in the second 48 hours went to ECMO whereas 2 of 9 not repaired went to ECMO. Multivariate analysis showed that timing of surgery is not a significant predictor of ECMO for patients with right CDH ($P = 0.399$).

Left Liver-Up

Thirteen of 21 (62%) patients repaired in the first 48 hours went to ECMO whereas 25 of 52 (48%) patients not repaired in the first 48 hours went to ECMO. Two of 7 (29%) patients repaired in the second 48 hours went to ECMO whereas 7 of 27 (26%) patients not repaired went to ECMO. Repair of CDH in the first or second 48 hours (Fig. 4) was not a significant predictor of ECMO by univariate or multivariate techniques ($P = 0.4330$).

DISCUSSION

The results of this large series, now matured over 19 years and with 268 consecutive patients, demonstrate several important findings. First, high survival rates were achieved using gentle ventilatory techniques and presently available treatment modalities. This finding is not new and has also been shown by others,[19–22] but the size of this experience and the magnitude of survival achieved make it important nonetheless. After removal of patients judged to have associated lethal anomalies, which represented 10% of the total, survival of patients with LLD CDH was 99%, right CDH was 90%, LLU CDH was 76%, and the aggregate survival in these patients was 88%. Survivors

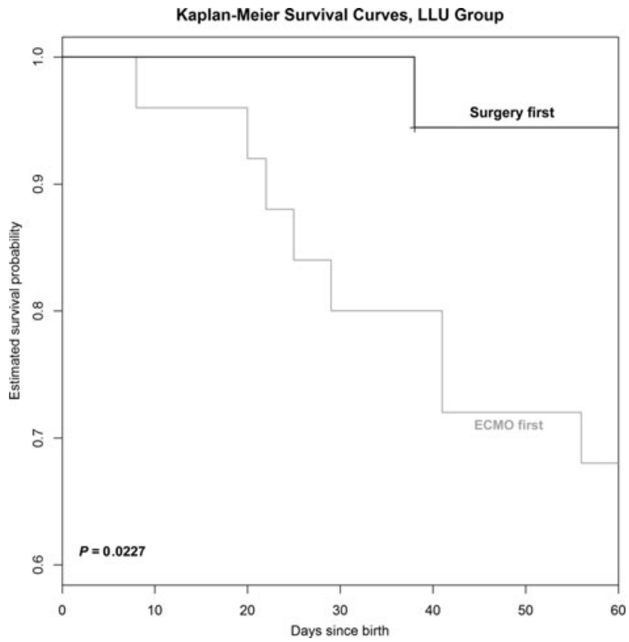

FIGURE 1. Kaplan-Meier survival curve comparing 18 patients with left liver-up CDH, who underwent repair before ECMO with 25 patients who went to ECMO unrepaired. ECMO indicates extracorporeal membrane oxygenation.

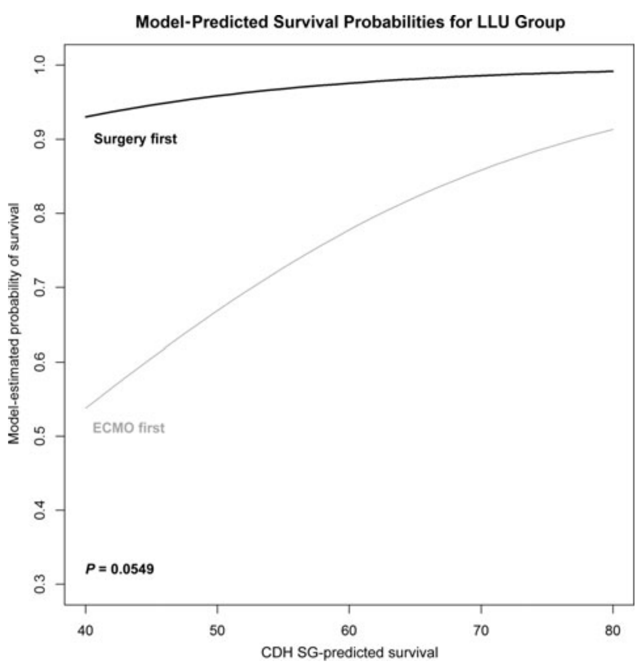

FIGURE 2. Multivariate logistic regression model comparing survival in 18 patients with left liver-up CDH who underwent repair before ECMO with 25 patients who had opportunity for repair but went to ECMO unrepaired, controlled for CDH SG–predicted survival. It is estimated that patients who are repaired first have 11.4 times the odds of survival as patients who arrive to ECMO unrepaired (95% CI = 1.41–286). The relationship does not quite reach statistical significance, $P = 0.0549$. CDH SG indicates CDH Study Group; ECMO, extracorporeal membrane oxygenation.

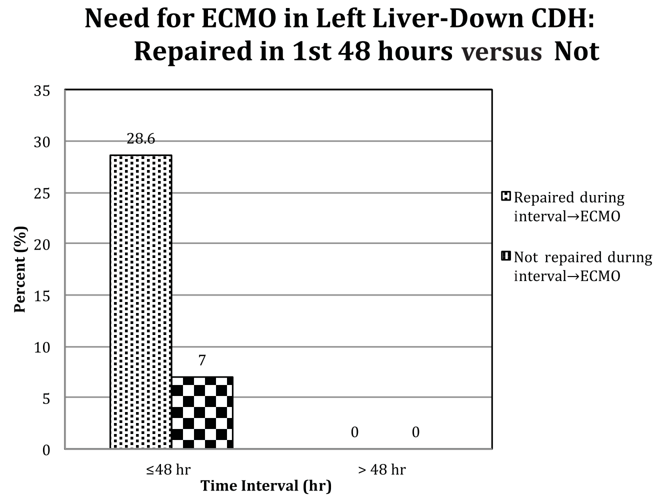

FIGURE 3. Timing of surgery is a significant predictor of ECMO in patients with CDH with left liver-down. Logistic regression controlled for predicted severity estimated that patients repaired within the first 48 hours have 6.3 times the odds of ECMO (95% CI = 1.55–28.1), $P = 0.0110$ compared with patients repaired later or not at all. ECMO indicates extracorporeal membrane oxygenation.

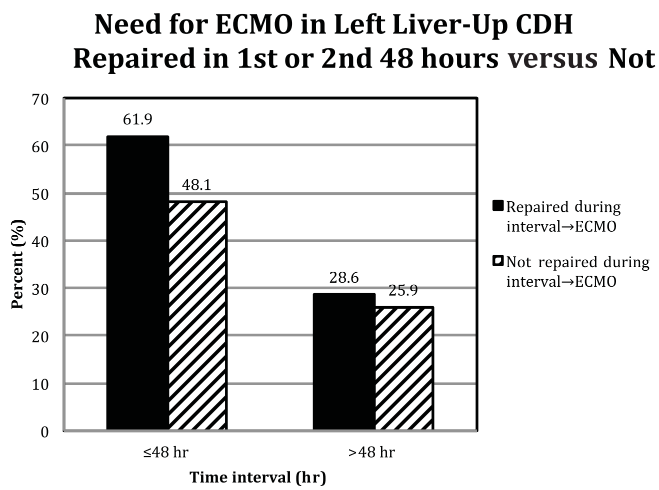

FIGURE 4. Timing of surgery is not a significant predictor of ECMO for LLU patients. $P = 0.4330$. ECMO indicates extracorporeal membrane oxygenation.

were discharged breathing spontaneously and on no additional respiratory support than supplemental oxygen, verifying that even patients with significant pulmonary hypoplasia can have good pulmonary outcomes. This information is important for physicians and prospective parents in evaluating treatment options and pregnancy termination.

The second important finding is that patients along the anatomic CDH spectrum are different and seem to benefit from different approaches to surgical timing. The data show with significant clarity that repair of CDH in the first 48 hours of life increases need for ECMO, and that this effect is most prominent in the least anatomically severe patients. Although the effect diminishes rapidly as CDH anatomic severity increases, these data show clearly that delay of repair beyond 48 hours is indicated in most patients to decrease subsequent need for ECMO.

The data also strongly suggest that patients on the severe end of the anatomic severity spectrum, those with LLU CDH, enjoy a survival benefit when repaired before ECMO. Although the multivariate analysis does not quite reach statistical significance, Figure 2 shows increasing divergence of the expected survival curves as predicted survival falls, supporting the postulate that patients with increasingly severe CDH benefit from a surgically more aggressive approach. This concept is new and may be counterintuitive for a generation of CDH physicians who have witnessed improvements in survival associated with delaying surgical repair. However, it may be that the main benefit in delaying repair is in decreasing need for ECMO, with resultant improved survival due to decreased need for ECMO.

The optimal use of the findings of this study requires refining our ability to define those LLU patients most likely to need ECMO. Examples of such predictive methods currently exist but will likely need to be center-specific for optimal utility.[23–25] Further application of prenatal imaging to quantify liver in the left side of the chest, as several authors have suggested, could also potentially aid in predicting risk for ECMO.[15,26,27]

Other methods for improving survival in patients with CDH who also require ECMO are described in the literature. A recent report from the CDH Study Group showed that repair after ECMO rather than repair on ECMO was associated with improved survival,[28] although the improved survival reported (∼65%) did not achieve the survival here (94%). Similarly, Dassinger et al[29] reported good results with early repair on ECMO (71%). Neither of these studies reflected a pure experience with LLU patients.

We see these data as representing an important step in understanding the differing effects of anatomic severity on CDH treatment effects and outcomes. Further work in this area is indicated and these data are expected help CDH physicians understand at a deeper level which patients benefit from surgical delay and which might benefit from earlier correction of surgical anatomy. Furthermore, these data should help in interpreting already published studies and in planning future studies by asking whether the data are controlled for anatomic severity and whether the conclusions apply equally to all patients across the CDH spectrum.

This study might be criticized for the relatively high utilization of ECMO. This rate is virtually identical to other contemporary studies.[5,21,29] Furthermore, we do not employ exclusion criteria from ECMO on the basis of disease severity, which may increase our use compared with others.

This report suffers from several weaknesses. The first is weakness inherent in a retrospective review, but this study is strengthened by its size and by the consistency of a single-institution experience with a highly consistent ventilation protocol and therapeutic oversight. Second, although there are strong trends in the data favoring a more aggressive surgical approach in patients with anatomically more severe CDH, the final multivariate analysis of the question posed here failed to meet accepted statistical significance, with a P value of 0.0549. Third, anatomic data such as LHR and volume of herniated liver, which could further subdivide anatomic severity in the LLU group, were lacking in enough patients over this 19-year experience that these data were not included. Finally, the clinical applicability of the observation that repair of CDH before ECMO seems to be beneficial in patients with severe CDH may be seen as providing little clinical guidance to bedside decision making, as the clinical decline in patients who subsequently require ECMO may be precipitous and leave little opportunity for safe repair.

CONCLUSIONS

In summary, these data show that patients with anatomically less severe CDH survive at high rates and have less risk of needing ECMO if repair is delayed beyond 48 hours. Patients with LLU CDH, however, seem to have a survival benefit when undergoing repair before ECMO. These data confirm the postulate that patients with CDH respond differently to treatment strategies across the anatomic severity spectrum and lay important groundwork for the development of a risk-specific treatment strategy for optimal management of CDH. Finally, the survival attained in this large and inclusive series of patients with CDH should be reassuring to physicians and parents faced with a new prenatal diagnosis of CDH.

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DISCUSSANTS

P. DILLON (Hershey, PA):

The authors present the largest single-institution retrospective CDH experience that has been published to date with truly outstanding clinical results. In 19 years, they have treated more than 260 patients with CDH and have a survival rate of 78%. When the infants with lethal anomalies were removed, their survival rate is 88%, and they have been able to maintain that over the 20 years. This is truly remarkable considering the registry data we have that is multi-institutional has the highest survival rate of 67%.

The core hypothesis that the authors attempt to address here is the timing of surgery in relation to the institution of ECMO therapy, and that is the area that I would like to focus on in my questions.

In terms of their results, they showed that the infants who had liver up, in other words, liver in the left side of the chest, above the diaphragmatic defect, had a greater chance of survival if surgery was performed early in the course of treatment before the initiation of ECMO, which runs a bit counter to current therapy, in which delayed therapy, sometimes as long as several weeks, has been proposed.

In the article, you mentioned at the very end the inborn/outborn status. I would be curious to know how that factored into your data analysis, at least in some type of multivariate regression, because I'm concerned that you may still have a selection bias in terms of how you ultimately arrived at surgical decision to manage those cases.

In the group that had surgical repair before or in anticipation of ECMO, how do you know that they truly needed ECMO? You did not have any standard criteria, I believe. And it was at the discretion of the surgical intervention that you chose to follow that pathway.

In the aggregate group, did you analyze your results over specific time periods? You put everything together into this data analysis. What would happen if you broke the process up into decade analysis or other time intervals to see whether there was indeed a change in your results over time?

Finally, in those patients who had surgical repair before ECMO, how do you explain your higher survival rate over the same category of patient in the CDH registry? What solid criteria can you propose for determining when surgery should be done before, during, or after ECMO?

Response From D.W. Kays:

The first question was about inborn versus outborn status. There was a tremendous increase in our inborn status. Multiple reports have shown that, in general, inborn patients are a higher-risk group and have lower survival than patients who are outborn. This might seem counterintuitive, but the sickest patients who are outborn die and never make it to management at the inborn hospital. This results in a selection bias as inborn patients are more ill and the fact that the survival was maintained with a tremendous increase in inborn patients adds to the power of the results.

How did we know whether the patients truly needed ECMO when we operated on them before ECMO? This is an important and difficult question that we addressed as best we could with comparisons of multiple variables. We compared the patients who needed ECMO with those who did not and showed significant differences. We then compared the patients who needed ECMO but were repaired first with those who went to ECMO first and did not find significant differences, showing that the patients who were treated with repair before ECMO were similar to the patients treated with ECMO first. You asked about the indications for ECMO, and I think I can best respond that it is our goal to never use ECMO, and we use all available modalities to what we consider are their therapeutic-toxic limits trying to avoid ECMO. ECMO is never instituted on the basis of risk of needing ECMO but only because of serious clinical declines affecting both pre- and postductal oxygenation. Going forward, it would be very valuable to develop robust methods and equations for predicting ECMO.

We did look at our own data to try to predict ECMO. We did logistic regression, starting with anatomic severity as the first variable, then adding lung-to-head ratio to the model. Although lung-to-head ratio was independently highly correlated with outcomes, the addition of that variable into the logistic regression equation did not give us acceptable sensitivity and specificity to predict ECMO well. We are continuing to analyze these data and are including more postnatal physiologic data such as CO_2 elimination and PO_2 levels extended to various time points to see whether we can improve our predictive abilities. We hope to report those results soon.

You asked a very interesting question about survival in different time eras over the 19-year experience. Although not presented here, the reason we started intervening more aggressively with surgical repair in sicker patients was the loss of a few patients in the middle of the experience who were treated with surgical delay. Our paradigm at that time was affected by national trends to delay surgery while focusing on pulmonary hypertension, but after losing these unrepaired patients on ECMO, patients we felt were otherwise completely survivable, we became more aggressive about getting the anatomy corrected as a key part of CDH treatment.

If we reported our results over 4- or 5-year time intervals, you would actually see that in the middle eras, our results worsened. I considered reporting this, but the numbers do not reach statistical significance over those intervals, only in aggregate. Because we have fully embraced a more aggressive surgical approach in the sickest patients, survival rates have recovered and exceed 90%.

You asked why survival in our patients exceeds that in the CDH registry. This is surely multifactorial and beyond the scope of this discussion, but we believe that each patient can survive and treat with that expectation. Also, our team benefits from consistent leadership with a clear therapeutic approach that covers all aspects of care including ventilation, fluids, repair, and ECMO. This results in a very high level of treatment consistency that, we believe, contributes to good outcomes.

DISCUSSANTS

D.P. Lund (Phoenix, AZ):

I rise to congratulate you on the contributions that the University of Florida, Gainesville, has made in the care of patients with diaphragmatic hernias. This has been one of the great problems for pediatric surgeons.

I have 2 questions.

The first is related to your ventilatory strategy over time. I was intrigued by the fact that your ECMO utilization rate did not really change between the first 60 patients you presented and the 200-plus you present here. It would strike me that if you had gone to a kinder, gentler ventilation strategy like most pediatric diaphragmatic hernia centers, your ECMO utilization rate would rise. So, I would ask you to comment on that.

The second question has to do with prenatal diagnosis and the advice that you give to parents who come to see you. The Holy Grail for in utero intervention has been to predict children who might need prenatal intervention on the basis of things such as lung-to-head ratio, liver-up, and so forth. Based on your experience, what advice are you giving to families that have potentially very severe congenital diaphragmatic hernia patients? And are you referring any of them to centers for consideration of in utero therapy?

Response From D.W. Kays:

Our ECMO utilization did not change significantly over the 2 periods. We were ventilating gently the whole time, and the ventilatory strategy really did not change. We did learn from this analysis that operating on babies on the milder end of the spectrum in the first 48 hours actually increases need for ECMO. We look forward to decreasing ECMO utilization in the future as we apply new lessons learned.

How do we deal with the prenatal evaluations? We believe, at a very fundamental level, that these babies have what it takes to survive and we treat each one as though he or she is going to survive. Although not presented here but described in the article, there were 44 patients from other states around the United States who traveled to the University of Florida Health Shands Children's Hospital for delivery and postnatal management. Of those 44 patients who did not have lethal or severe associated anomalies, 42 went home with their families, for a 95.5% survival rate. In short, we are very optimistic about patients with CDH treated at our institution and at this point do not refer them out for prenatal interventions. We inform them of our results and let them make their own decisions.

DISCUSSANTS

R.J. Touloukian (New Haven, CT):

My question is about selecting the size and the position of the liver as the best surrogate for survival. As you know, there are many studies that use other parameters. You have mentioned lung development, certainly. But there are others, including the progression of lung size over time, the presence of the stomach above or below the diaphragm, and the issue of whether pneumothorax was present at the time of birth. What percentages of patients were inborn and whether you tested liver-up and liver-down against any of the other standard surrogates for survival?

Response From D.W. Kays:

We actually started our analysis with a 5-step definition of anatomic severity, 1 right-sided and 4 left-sided. We looked at left liver-down, stomach-down; we looked at liver-down, stomach-up. We looked at liver-up, small amount of liver and then liver-up, large amount of liver to give us 4 left-sided groups. Because the numbers were small relative to left-sided, we did not look at subgroups of right-sided CDH. A great deal of effort went into analyzing all of these subgroups.

It turned out that what really mattered in our series was liver-down versus liver-up, and subdefinitions added little to the analysis. Survival in patients who were liver-down, whether it was stomach-up or stomach-down, was 99%, and adding the variable stomach-down or stomach-up was not useful. To define the amount of liver-up, we noted the position of the falciform ligament at the time of surgery or autopsy, with liver-up to falciform ligament defined as "small" liver-up and greater than this being "large" liver-up. But in a series that spanned 19 years, we did not have that level of definition in all patients and felt that assigning this without clear data risked adding bias, so we removed that subgrouping.

In the end, it turned out that the analysis was cleaner and more meaningful using the simple but powerful variable of liver-down or liver-up.

The other reason we favored analysis by anatomic rather than lung measurement data was that we had this level of anatomic data on every patient, whereas we did not have lung-to-head ratio data for all of our patients. Especially early in the series, many of the patients were outborn and had no prenatal data.

Regarding pneumothorax as a defining variable, our pneumothorax rate is less than 1% and, therefore, is not an adequate discriminator. But there is no doubt that other grading systems can be used, and modern magnetic resonance imaging analysis can further refine liver-up severity.

A final message I would like to reiterate with this group of esteemed surgeons is that congenital diaphragmatic hernia is a surgical disease. The survival of unrepaired patients, except in those minimally affected, is zero. We greatly appreciate the opportunity to present these data and thank you for your attention and questions.